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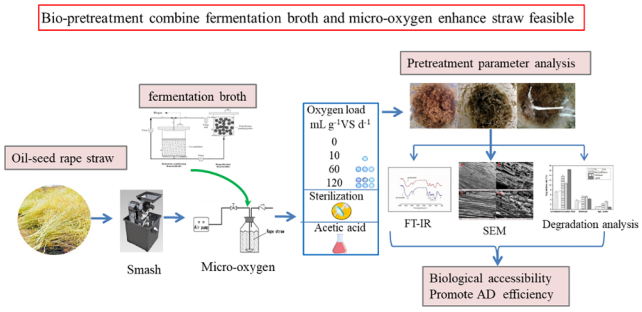
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Bio-pretreatment promote hydrolysis and acidification of oilseed rape straw: Roles of fermentation broth and micro-oxygen

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GRAPHICAL ABSTRACT



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ABSTRACT

Oilseed rape straw (ORS) is capable of producing renewable energy. However, cellulose, hemicellulose and lignin are intertwined together in ORS, which makes it difficult for anaerobic digestion (AD). Hence, pretreatment is the key factor in reducing the rate-limiting step of AD. This study reports that the pretreatment combined fermentation broth and micro-oxygen could enhance the degradation of ORS. The maximum biodegradation ratios of cellulose, hemicellulose, and lignin (CHL) were 20.6%, 18.1%, and 24.7%, respectively, at 120 mL/gVS/d oxygen load. The maximum volatile fatty acids and soluble chemical oxygen demand of hydrolysis and acidification of the pretreated groups were significantly higher than that of the control groups. Microorganisms in the fermentation broth at micro-aerobic conditions led to the reduction of CHL content, and altered the structure of ORS. The fermentation broth bio-pretreatment could effectively decrease the functional groups related to lignin.

1. Introduction

In China, oilseed rape straw (ORS) (kind of dicotyledonous crop) is one of the widely cultivated oil-crops. More than 20 million tons of ORS are produced each year (Wang et al., 2017a,b). The ORS has been recognized as a source of renewable energy. Anaerobic digestion (AD) is known as a widely used and cost-effective way to convert agriculture biomass into biogas (Sawatdeenarunat et al., 2015; Wu et al., 2019).

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However, cellulose, hemicellulose and lignin are intertwined with each other and forming complex structures in ORS, which makes it difficult to be decomposed by the microorganisms during the AD process (Patinvoh et al., 2017).

It is recognized that the rate-limiting step of AD mediates the initial hydrolysis process of macromolecular organic matter or non-degraded organics, especially in the two-stage anaerobic digestion. Hence, pretreatment of the straw before hydrolysis has been considered as an essential step (Tian et al., 2012). Thus, several efforts have been made to enhance the efficiency of AD, such as thermal pretreatment (Luo et al., 2019; Phuttaro et al., 2019), thermal-chemical pretreatment mechanical pretreatment (Qiao et al., 2018; Vivekanand et al., 2012), and biological pretreatment (Guan et al., 2018; Guo et al., 2019). In addition, steam explosion has been reported as an effective method that resulted in the transformation of hemicellulose under explosive decompression (Wood et al., 2014). The alkali pretreatment is also an effective and commonly used way to selectively remove lignin from lignocelluloses (Sun et al., 2015). However, the extra energy or chemicals are often required in these physicochemical treatment methods, which leads to the increased cost and difficulties in recovering the chemicals. Thus, these physic-chemical pretreatments have rarely been applied in a full-scale plant (Carrere et al., 2016). Compared with physical and chemical pretreatment methods, biological pretreatment is a relatively inexpensive and low energy demand upstream technology.

Up to now, extensive studies have been done on biological pretreatment of lignocellulose, such as enzymatic, fungi, composting and ensiling pretreatments (Xu et al., 2018). However, it might be difficult to widely application in biogas plants due to the cost of microbial strains and enzymes (Carrere et al., 2016). In addition, N element was added to adjust the C/N ratio in white-rot fungi biological pretreatment inoculants (Castoldi et al., 2014). In the actual biogas production process, a large amount of biogas slurry and biogas residue would be produced. More importantly, the biogas slurry usually contained a high concentration of ammonia nitrogen that can cause water pollution (Chen et al., 2017). However, the treatment of ammonia nitrogen in biogas slurry focuses on the recovery of additional chemicals, which results in a relatively high cost. The pretreatment of the straw with the fermentation broth containing microorganisms can be a promising pretreatment inoculum, and the nitrogen element (NH_4^+- N) can be used as nutrition for the microorganism grow (Guo et al., 2019).

A previous study found that aerobic pretreatment could lead to low volatile fatty acids (VFAs) concentration and high alkalinity. However, methane production from per unit of volatile solids (VS) was not improved (Ward et al., 1998). In addition, the micro-aerobic condition in anaerobic digester by the addition of a low oxygen dose could significantly increase the hydrolysis rate of lignocellulosic (Fu et al., 2016) and methane production in the anaerobic process (Ramos and Fdz-Polanco, 2013; Xu et al., 2018). Up to now, it is unclear whether ORS could promote the AD ability with fermentation broth under micro-aerobic condition. The objective of this study is to investigate the effects of the bio-pretreatment on VFAs production and several anaerobic fermentation parameters.

2. Materials and methods

2.1. Raw materials

The ORS used in this study was collected from Hexian District, Maanshan City, Anhui Province, China. The ORS was laid over the field and dry under the sun for 1 week, and the moisture content was about 10%. The contents of C and N in ORS were 44.51 ± 0.32% and 0.88 ± 0.11%, respectively, TS and VS were 91.21 ± 0.66% and 86.63 ± 0.53%, respectively. The cellulose, hemicellulose and lignin contents were 40.64 ± 0.92%, 23.15 ± 0.87% and 21.33 ± 0.54%, respectively. The pretreatment liquid used for the pretreatment was taken from the hydrolysis acidification reactor of the two-phase anaerobic reactor, which was normally operated in the laboratory. The substrate was a mixture of rapeseed straw, kitchen waste and duck feces with a dry matter ratio of 2:1:1 (Wang et al., 2017a,b). The pH of the pretreatment liquid was 6.81 ± 0.16. sCOD concentration, VFAs concentration and ammonia nitrogen concentration were 0.78 ± 0.11 g/L, 0.45 ± 0.08 g/L and 3.2 ± 0.18 g/L, respectively. The microbial community structure analysis of pretreatment liquid showed that there were a total of 16 identified bacterial phyla and a total of 7 identified fungal class. The most dominant bacterial phyla was *Bacteroidetes* (50.26%), and followed by *Firmicutes* (15.37%) and *Proteobacteria* (9.34%). The most dominant fungal class was *Sordariomycetes* (44.99%), and followed by *Dothideomycetes* (25.56%) and *Eurotiomycetes* (17.9%).

2.2. Biological micro-aerobic pretreatment

Micro-aerobic pretreatment was applied to treat the ORS upstream phased anaerobic digestion. Before pretreatment, the 500 mL wide-mouth jars were autoclaved for 20 min at 121 °C. Every 30 g of straw was placed in a 500 mL wide-mouth jar, evenly sprayed with 58 mL of fermentation broth from phased anaerobic digester (hydrolysis acidification reactor) and evenly stirred with Titanium stick. The pretreatment moisture content was 65%, and the pretreatment time was 10 d. The jars were sealed with perforated rubber stoppers. Micro air pumps (1 L/min) were used to ventilate the aeration head installed in the middle of the bottle bottom. The oxygen loads were 0, 20, 60 and 120 mL/gVS/d, corresponding to T0, T1, T2 and T3 test. The ORS in the control group used deionized water instead of fermentation broth.

In order to investigate the role of microorganisms and other components (eg. organic acids), the hydrolysis acidification liquid was treated as follows: in T4 test, the liquid was filtered through 0.22 μm microporous membrane (33 mm/0.22 μm, Millipore, USA) and used as a sterilization pretreatment liquid. The oxygen load of T4 was 20 mL/gVS/d (optimal oxygen load in previous laboratory experiments).

After the pretreatment, the solid residue was filtered and washed several times by deionized water until neutral. The collected dry solids were used for hydrolysis and acidification tests and other analyses. The experiments were carried out in triplicates for each group. At the end of the pre-treatment, the straw samples were taken out to determine CHL (cellulose, hemicellulose and lignin) content and dry weight. The degradation ratio of CHL of pretreated ORS was calculated as Eq. (1).

$$\text{CHL degradation ratio}\,(\%) = (W_1 - W_2)/W_1 \times 100\% \quad (1)$$

W1- The amount of CHL in untreated ORS (previous content % × total weight × TS %).

W2- The amount of CHL in the ORS after pretreatment (content after treatment % × total weight × TS %).

In order to investigate the important role of pretreatment liquid in the process of straw pretreatment, the contribution rate of microorganisms in the pretreatment process was calculated as Eq. (2) (Tian et al., 2017).

$$\text{Microbial contribution rate} = \frac{\text{CHL degradation ratio in pretreatment liquid group} - \text{CHL reduction ratio in sterilization group}}{\text{CHL degradation ratio in pretreatment liquid group}} \times 100\% \quad (2)$$

2.3. Hydrolysis and acidification test

Pretreated ORS or untreated ORS were mixed with inoculum (taken from laboratory hydrolysis acidification reactor) in a 250 mL serum bottles. The working volume and the total TS loading were controlled at 200 mL and 60 g TS/L, respectively.

All serum bottles were tightly closed with butyl rubber stoppers and aluminum-plastics combined caps. No additional nutrient solution was added to anaerobic hydrolysis and acidification tests. All the serum bottles were placed in constant temperature shaker incubator (Xinrui, THZ-82, China) at 35 °C and shaken 8 times per day at 100 r/min for 5 min. The hydrolytic acidification solution was sampled daily to determine the sCOD and VFAs, and the remaining liquid after the analysis was returned to the reactor with a syringe (Mohan and Bindhu, 2008). In order to eliminate the influence of VFAs in the inoculum on the concentration of sCOD and VFAs during hydrolysis and acidification, the amount of sCOD and VFAs brought into the inoculum was subtracted when calculating the concentration.

2.4. Analytical methods

All experiments were performed in duplicate. The analyses of total solids (TS), volatile solids (VS) were conducted in accordance with Standard Methods (APHA, 1998). Cellulose, hemicellulose, and lignin content were determined according to the method described by Van Soest (Van Soest et al., 1991). sCOD and VFAs in liquid samples were analyzed after particulate removal by centrifuged at 10,000 rpm for 10 min and filtered through 0.45 μm membrane (33 mm/0.45 μm, Millipore, USA). The concentration of VFAs in hydrolytic acidification fluid was measured by GC (GC-2001, Shimadzu, Japan) equipped with hydrogen flame ionization detector (FID), using quartz capillary column (30 m × 0.32 mm × D × 0.25 μm; Stabilwax-DA, RESTEK, USA) with nitrogen as a carrier gas. GC operating conditions were as follows: injector temperature at 220 °C, detector temperature at 230 °C. Column heating procedures were as follows: the starting temperature was maintained at 100 °C for 1 min, then at 10 °C/min rate increased to 140 °C and maintained 2 min, at last increased to 200 °C at 15 °C/min rate and maintained 2 min.

2.5. Data analyses and statistics

The measurements were expressed as means ± standard deviation. The significant differences between groups were determined using one-way analysis of variance. The significant level (p value) was set at 0.05 and 0.01. Pair-wise comparisons were conducted on sCOD concentration, VFAs concentration and the CHL degradation ratio to identify significant differences between different groups. The Pearson correlation analysis was used to analyze the correlation between experimental data. The SPSS19.0 (IBM, USA) and OriginPro 8.5 (OriginLab, USA) were used for data processing and graphing.

2.6. Chemical and physical analysis

In order to intuitively explore the changes of surface morphology and internal structure of rapeseed straw under micro-aerobic pretreatment, field emission scanning electron microscope (FESEM, Ultra Plus, Zeiss, Germany) was used to analyze the untreated straw and pretreated straw samples. The specific steps were as fellow: taking an appropriate amount of untreated or pretreated straw granules first and then vacuum drying with a freeze dryer (FD-1A-80, Tianling, China). The dried samples were fixed on the silicon wafer and placed in the sample chamber for observation at 1.0 kV.

Fourier transform infrared spectroscopy (FTIR, Nicolet iS50, Thermo Fisher Scientific, USA) was used to investigate organic functional groups and structural changes in ORS. The specific steps were as fellow: (1) weighed 1 mg of the ORS sample after freeze-drying, (2) added 100 mg of dried potassium bromide powder, (3) mixed well, (4) ground in an agate mortar and putted into a molding machine, (5) vacuumed to increase pressure and made into a transparent tablet under pressure. The pellets were placed in a Fourier transform infrared spectrometer (iS50, Thermo Scientific Nicolet, USA) with a resolution of 4 cm^{-1} and a number of scans of 30.

X-ray diffraction (XRD, SmartLab3, Rigaku, Japan) was used to investigate crystalline structure changes in ORS, and use diffraction intensity to calculate the crystallinity index of cellulose (Zhao et al., 2016a,b). The specific steps were as fellow: grinded the ORS sample with agate mortar after freeze-drying, and putted the sifted ORS sample in X-ray diffractometer (Smartlab (3), Rigaku Corporation, Japan) for measurement. XRD operating conditions were as follows: Cu target, Ni filter, tube current was 30 mA, tube voltage was 40 KV, scan mode was $2\theta/\theta$ scan, the measuring range of angle was 10–60°, scan rate was 5°/min.

3. Results and discussion

3.1. Micro-oxygen increase bio-pretreatment of ORS

To measure the effects of fermentation broth and micro-oxygen on the degradation of ORS, the biodegradation rates of cellulose, hemicellulose, and lignin were evaluated. Cellulose and hemicellulose in ORS are the main carbon sources during the anaerobic process (Zhong et al., 2011). Changes in cellulose, hemicellulose, lignin, TS and VS of ORS after the pretreatment process were shown in Table 1. After the pretreatment, both TS and VS were reduced with the increase of oxygen load. The lignin and cellulose content were reduced with the oxygen load increased, and the content of hemicellulose were remained stable. Though decreasing lignin content could improve the biological accessibility of treated ORS, the cellulose content was also dropped down.

Oxygen supply is one of the particularly important factors for the ORS pretreatment. However, excess oxygen would lead to the massive proliferation of aerobic microorganisms, which result in the easily degradable COD directly oxidized to H_2O and CO_2 during micro-aerobic pretreatment (Zitomer and Shrout, 1998). The degradation ratios of

Table 1
Content (%) of main composition for ORS after pretreatment process.

	Cellulose	Hemicellulose	Lignin	TS	VS
Untreated	40.64 ± 0.92	23.15 ± 0.87	21.33 ± 0.54	91.21 ± 0.66	86.63 ± 0.53
T0	39.85 ± 0.75	22.32 ± 0.78	22.05 ± 0.61	88.24 ± 0.55	82.64 ± 0.57
T1	39.77 ± 0.81	21.83 ± 0.65	19.53 ± 0.56	88.05 ± 0.59	82.37 ± 0.45
T2	38.62 ± 0.79	22.04 ± 0.71	18.59 ± 0.51	87.82 ± 0.44	81.59 ± 0.58
T3	38.13 ± 0.84	22.42 ± 0.79	18.97 ± 0.60	85.25 ± 0.63	78.63 ± 0.41
T4	40.34 ± 0.85	22.89 ± 0.82	21.37 ± 0.47	89.13 ± 0.45	84.89 ± 0.65

The oxygen loads of T0, T1, T2 and T3 were 0, 20, 60 and 120 mL/gVS/d, respectively. The oxygen load of T4 was 20 mL/gVS/d with sterilization pretreatment liquid. Data are the averages of three measurements, ± standard error of three replicates.

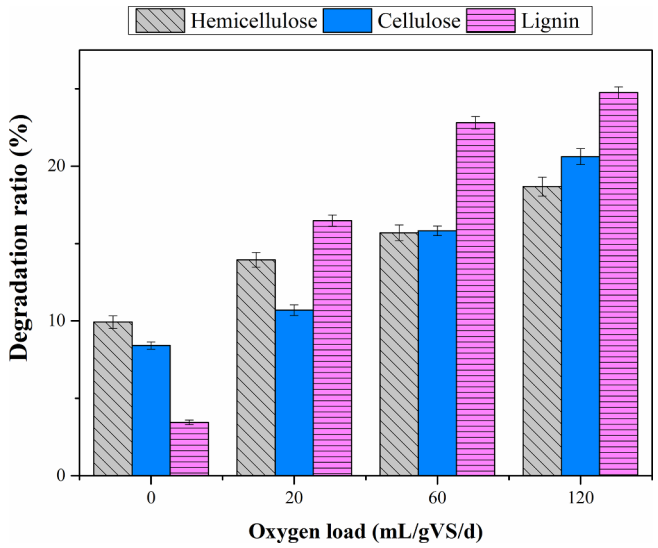


Fig. 1. Oxygen load effect on CHL degradation of ORS.

lignin, hemicellulose and cellulose of ORS under micro-aerobic bio-pretreatments were shown in Fig. 1.

As shown in Fig. 1, the degradation ratio of CHL increased with the increase of oxygen load, and the degradation of lignin was most significant. When the oxygen load was 120 mL/gVS/d, the degradation ratio of lignin reached 24.74% after 10 days, and the highest degradation ratios of cellulose and hemicellulose were 20.61% and 18.05%, respectively. This was due to the fact that most of the hemicellulose and cellulose in the straw were encapsulated by lignin, and the degradation of lignin during pretreatment could increase the chances of hemicellulose and cellulose in the interior being in contact with microorganisms (Zhao et al., 2016a,b). Most of the actinomycetes and molds capable of degrading lignin were obligate aerobic bacteria, which used molecular oxygen as the final electron (hydrogen) receptor in the respiratory process. These microorganisms will be more active and consumed more organics with the increasing oxygen load (Díaz et al., 2011).

It's worth noting that the degradation ratio of cellulose, hemicellulose and lignin were 21.02%, 18.58% and 25.03%, respectively, when oxygen load increased to 150 mL/gVS/d. The degradation ratio of CHL were not significantly higher ($p > 0.05$) than that in T3 group. This indicates that the biodegradability of bio-pretreatment is hard to increase further when oxygen load higher than 120 mL/gVS/d. The CHL degradation in different pretreatments of some agro-industrial wastes were summarized in Table 2. This study showed higher cellulose and lignin degradation than that of traditional pretreatment (Zhou et al., 2012). The lignin and hemicellulose degradation of ORS in this study were similar to those of fungal pretreatment. However, fungal pretreatment has better selectivity in lignin removal, and the cellulose degradation in this study was higher than that of fungal pretreatment (Ge et al., 2015). The degradation profile of ORS was also similar to rice straw pretreated by combined biological and chemical pretreatment,

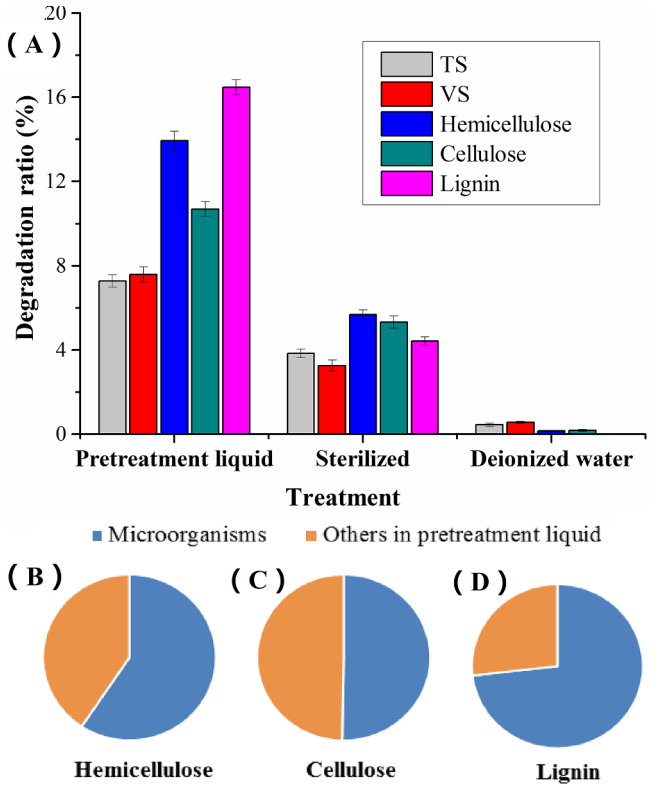


Fig. 2. Degradation performance of ORS pretreated by fermentation broth. A, Degradation of VS, TS and CHL of rape straw pretreated by fermentation broth and deionized water. Contribution of microorganisms and other components for the degradation ratios of hemicellulose (B), cellulose (C), and lignin (D).

and the chemicals greatly reduce the treating time than bio-pretreatment (Guan et al., 2018).

3.2. Pretreatment liquid enhances the degradation of ORS

The pretreatment liquid (fermentation broth) could increase the degradation performance of TS, VS and CHL (Fig. 2A). For instance, the degradation ratios of TS and VS for the ORS in the original inoculum group were 7.27% and 7.59%, respectively. In addition, the degradation ratios of cellulose, hemicellulose, and lignin were 10.71%, 13.90%, and 16.50%, respectively. These degradation ratios of VS, TS and CHL in the T1 group (pretreatment liquid group) were significantly higher ($p < 0.05$) than that of the control group (deionized water). The degradation ratio of the control group showed that the loss of VS, TS and CHL were all lower than 0.5%. This indicated that the waxy surfaces and lignocellulose structure of ORS could prevent organic matter dissolved into water.

Previous studies have suggested that pretreatment liquid-increased degradation ratios of straw might due to the combined effect of microorganisms and other components (organic acids) (Amnuaycheewa

Table 2
Degradation of cellulose, hemicellulose and lignin in different pretreatments.

	Waste	Pretreatment conditions	Degradation ratio (%)		
			Cellulose	Hemicellulose	Lignin
T2	Oilseed rape straw	10d, room temperature	15.82	15.68	22.80
Biological and chemical pretreatment (Guan et al., 2018)	Rice straw	3d, room temperature	1.08–16.19	6.18–25.79	6.20–22.45
Fungal pretreatment (Ge et al., 2015)	Albizia Chips	48d, 28 °C	10.50	15.00	24.00
Stack-pretreatment (Zhou et al., 2012)	Corn stover	20d, room temperature	5.80	16.80	5.70

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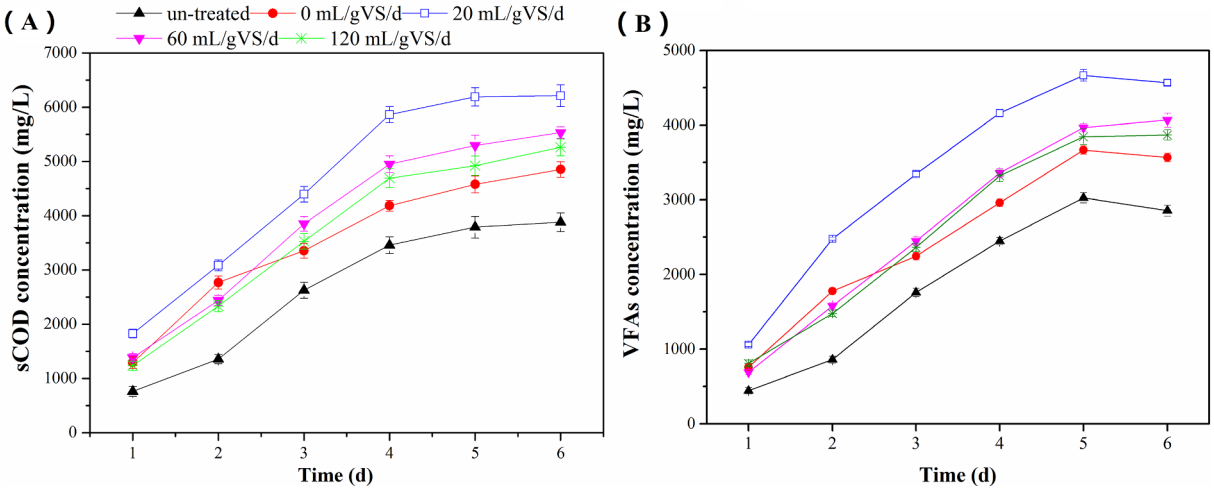


Fig. 3. Hydrolytic acidification of pretreated ORS under different oxygen load. A, changes of sCOD; B, changes of VFAs.

et al., 2016). In this study, the effects of sterilizing fermentation broth on the degradation performance of hemicellulose, cellulose, and lignin for ORS under micro-aerobic conditions were obtained (Fig. 3A). The result indicated that the degradation amount of organic matter (TS, VS) and CHL in the pretreatment process were greatly reduced after the removal of microorganisms from pretreatment liquid (sterilized group). In order to further verify the above phenomenon, the contribution rates of microorganisms to cellulose, hemicellulose and lignin degradation were 59.3%, 50.2% and 73.2%, respectively in the T1 group (. 3B, C, and D, calculated by eq2). It was obvious that the microorganisms in the pretreatment liquid played an important role in the pretreatment process. In addition, the degradation ratios of TS, VS and CHL for the ORS pretreat by sterilized pretreatment liquid were higher than that in the control group. It was mainly due to those other organic acids, such as acetic acid, propionic acid, butyric acid, and valeric acid, etc., in the pretreatment liquid, which can remove lignin, hemicellulose and cellulose in straw under normal pressure and mild conditions (Kivaisi and Eliapenda, 1995).

3.3. Performance of hydrolysis and acidification for the bio-pretreatment ORS

Previous research found that the sCOD was positively correlated with methane yields during AD (Shen et al., 2014). The hydrolytic-acidification experiment was investigated to explore the performance of hydrolysis and acidification in the bio-pretreated ORS.

The trend of sCOD and VFAs were analyzed during the operation period. For example, the sCOD of each group continued to rise in the first 4 days, and tended to be gentle, reached 3789.0–6192.6 mg/L on the 5th day, and then decreased slightly on the 6th day. While the VFAs continued to rise to 3025.3–4667.3 mg/L in the first 5 days, and decreased slightly from the 6th day. This indicated that under the hydrolysis of microorganisms in the first 4 days, cellulose, hemicellulose and other components in the ORS were hydrolyzed into soluble substances, and then the soluble substances were converted into volatile organic acids by the action of acidified fermentation bacteria. Combined with the trend of VFAs and sCOD, the hydrolysis acidification peaked on the fifth day, after which the VFAs began to be consumed.

The performance of hydrolysis and acidification for the bio-pretreatment ORS was higher than that in the control group (without bio-pretreated ORS). For instance, the highest VFAs concentrations in the pretreatment group with oxygen loads of 0, 20, 60 and 120 mL/gVS/d were 21.2%, 54.3%, 31.1% and 27.1% higher than the untreated group, respectively. The highest concentrations of VFAs and sCOD were both obtained at 20 mL/gVS/d oxygen load group, which reaching 4667.3 ± 76.7 mg/L and 6192.6 ± 167.2 mg/L on the fifth day, respectively. The result was different from the degradation ratio changes of lignin increased with the increase of oxygen load in Section 3.1. This was mainly because the increase in oxygen supply would proliferate a large number of aerobic bacteria and increase the degradation of lignin. However, the intermediate products produced by the pretreatment process of cellulose, hemicellulose and other components in ORS would be directly mineralized, thereby reduced the available components in the subsequent hydrolysis acidification process (Koch et al., 2015).

3.4. Characteristics of bio-pretreated ORS

3.4.1. Apparent morphology of ORS

As shown in the image during pretreatment, white hyphae gradually covered the surface of the straw particles with the micro-aerobic pretreatment process, and the mycelium was uniformly distributed on the surface of the straw particles at 10 days. After 25 days, the mycelium grew into a bud. The photographs by optical microscope before and after pretreatment shows the fiber bundles inside the untreated straw were clearly visible, and the fibers are arranged closely and regularly. From the optical micrograph of the pretreated straw, it can be seen that there were obvious hyphae in the straw. These hyphae extending into the interior could play a certain destructive effect on the initially arranged fiber tube so that more cellulose and hemicellulose could be exposed, and the treated ORS became more accessible to hydrolytic/acidification microorganism.

It can be seen from FESEM images that the surface structure of the straw was tight and regular without the pretreatment of hydrolytic acidification fluid and micro-aerobic, and there was almost no gap between the fiber bundles. However, after the pretreatment, the surface structure of the straw changed significantly. A large number of holes were formed on the surface of the treated straw, and the structure became disordered. The individual fiber bundles produced a spring-like spiral structure under the microbial erosion. FESEM images shown that microorganisms played an important role in the microaerobic pretreatment process. A large number of microorganisms were distributed on the surface and inside of the straw samples. According to the morphology, there were bacteria, actinomycetes and fungi. These microorganisms could secrete various enzymes to degrade the polysaccharides, proteins, lignin, and cellulose. The mycelium of actinomycetes and fungi could penetrate the surface of the dense lignocellulose, deep into the interior and degrade the straw while causing damage to its structure (Guan et al., 2018).

3.4.2. FTIR analysis

The components in the ORS are composed of some basic structural units and these units have specific chemical groups. Therefore, changes in the composition of ORS before and after pretreatment were analyzed by FTIR. As shown in infrared spectra, the symmetrical or anti-symmetric stretching vibration of –CH3, –CH2 and aliphatic CH near 2920/cm was weakened after the micro-aerobic biological pretreatment. It can be speculated that this was related to the breakage of some fatty chains in oilseed rape straw and the degradation of some carbohydrates. After pretreatment, the C═O stretching vibration associated with the aromatic ring near 1636–1634/cm was significantly weakened, which indicated that the lignin in the straw was degraded. The absorption peaks around 1509, 1424, and 1326/cm were also related to lignin, which was related to the benzene ring, the methyl group C–H, and the lilac ring. The three absorption peaks decreased slightly after pretreatment, which was consistent with the lignin degradation in the ORS as shown in Fig. 1.

The peak intensity of the straw after pretreatment at 1245/cm was much weaker than that of the untreated straw, which was related to the microbial degradation of the guaiac aromatic ring in the pretreatment process (Zhao et al., 2016a,b). The absorption peak near 1066/cm was related to the vibration of the C-O extension peak of cellulose and hemicellulose, and was also related to the stretching vibration of siloxy group in silica (O-Si-O) (Zhao et al., 2016a,b). The spectra showed that the spikes near 1066/cm were wider and more gradual after pretreatment, which indicated that the crystalline silicon and cellulose in the rape straw had been transformed (Fu et al., 2015a,b). When wheat straw was treated with NaOH, the absorption peak near 1066 disappeared, mainly because NaOH can break the ester bond between carbohydrate and lignin, thereby released the fibers wrapped in lignin (Fu et al., 2015a,b).

3.4.3. XRD analysis

In the aspect of straw against biodegradation, the crystalline cellulose is harder accessible to cellulase attack than amorphous cellulose has been widely accepted. It means biodegradation of ORS is inverse to the amount of crystalline cellulose. Therefore, changes in crystallinity of ORS before and after pretreatment were analyzed by XRD. After biological micro-aerobic pretreatment, the main broad peak at 22° and peak at 16.2°, which corresponding to the crystalline portion and the amorphous portion of ORS (Dhabhai et al., 2013). The calculated crystallinity indexes of untreated and pretreated ORS were 39.25% and 35.11%, respectively. This indicates that biological micro-aerobic pretreatment modified the crystallographic structure of rape straw, and reduced the amount of crystalline cellulose present in the rape straw. Hence the improvement of hydrolysis and acidification efficiency of ORS can be attributed to the crystalline cellulose reduced in the pretreatment process.

4. Conclusions

This study proved that fermentation broth could enhance the biodegradation of ORS under micro-aerobic conditions. Results showed that CHL degradation of ORS can be increased with oxygen load increased from 0 to 120 mL/gVS/d. The maximum concentrations of VFAs and sCOD were obtained at 20 mL/gVS/d oxygen load. Microorganisms in fermentation broth played critical roles during the pretreatment process. The physical and chemical characterizations of bio-pretreated ORS showed that the destruction of functional groups related to lignin and reduction of cellulose crystallinity index might be the reason for the improvement of hydrolysis and acidification.

CRediT authorship contribution statement

Chuqiao Wang: Resources, Validation, Formal analysis, Visualization, Software, Writing - original draft, Data curation. **Jiajie Zhang:** Validation, Formal analysis, Visualization. **Fengping Hu:** Writing - review & editing. **Shuai Zhang:** Conceptualization, Methodology, Software, Investigation, Writing - review & editing. **Ji Lu:** . **Shuai Liu:** Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2020.123272.

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